2. ROS2 install Jazzy

- ROS2 Jazzy corresponds to Ubuntu 24.04; this tutorial installs ROS2 Jazzy on Ubuntu 24.04.
- The software sources, packages and interfaces may differ between Ubuntu and ROS2 versions, so please use the ROS2 version that matches your system, and do not simply replace the version name and run the commands directly.
- The factory system already comes with the jazzy ROS environment set up; this tutorial applies to pure systems.

1. Setting the language environment

First check whether the local language environment supports UTF-8 encoding. You can check it and set UTF-8 encoding with the following commands:

```
locale  # check whether UTF-8 is supported

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale  # verify that the setting worked
```

Note: the language environment can be different, but it must support UTF-8 encoding.

2. Setting the software source

Enable the Ubuntu universe repository

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

Install the ROS 2 software source configuration package. This method automatically configures the GPG key and the software source that matches the current Ubuntu version:

```
sudo apt update && sudo apt install curl -y
export ROS_APT_SOURCE_VERSION=$(curl -s https://api.github.com/repos/ros-infrastructure/ros-apt-source/releases/latest | grep -F 'tag_name' | awk -F\" '{print $4}')
curl -L -o /tmp/ros2-apt-source.deb "https://github.com/ros-infrastructure/ros-apt-source/releases/download/${ROS_APT_SOURCE_VERSION}/ros2-apt-source_${ROS_APT_SOURCE_VERSION}.$(. /etc/os-release && echo ${VERSION_CODENAME})_all.deb"
sudo dpkg -i /tmp/ros2-apt-source.deb
rm -f /tmp/ros2-apt-source.deb
```

3. Installing Jazzy

First update the apt repository cache:

```
sudo apt update
```

Then upgrade the installed software (the ROS2 packages are built on an Ubuntu system that is updated frequently, so please make sure your system is up to date before installing new packages):

```
sudo apt upgrade
```

Install the desktop version of ROS2 (recommended), which includes ROS, RViz2, the examples and the tutorials:

```
sudo apt install ros-jazzy-desktop python3-argcomplete
```

Install the colcon build tool

```
sudo apt install python3-colcon-common-extensions
```

4. Configuring the environment

In the terminal, when you run a ROS2 program, you have to call the following command to configure the environment:

```
source /opt/ros/jazzy/setup.bash
```

Every time you open a new terminal you have to run the command above, or you can run the following command to write the environment configuration command into the "~/.bashrc" file, so that you do not have to configure the environment manually every time you start a new terminal

```
echo '[ -f /opt/ros/jazzy/setup.bash ] && source /opt/ros/jazzy/setup.bash' >> ~/.bashrc
```

After reopening the terminal, check the environment with the following command:

```
echo $ROS_DISTRO
command -v ros2
```

You should see `jazzy` and `/opt/ros/jazzy/bin/ros2` respectively. If you additionally set `PYTHONPATH`, do not overwrite the Python package search path of ROS2; it is recommended to use the append form, for example:

```
export PYTHONPATH="/home/jetson/jetcam${PYTHONPATH:+:$PYTHONPATH}"
```

At this point ROS2 has been installed and configured.

5. Running the example publisher and subscriber nodes

The ros2 desktop version integrates some example nodes for us, and we can run the example nodes to test whether ros2 has been installed successfully.

- Run the publisher node

```
ros2 run demo_nodes_cpp talker
```

- Run the subscriber node

```
ros2 run demo_nodes_cpp listener
```

6. Running the little turtle example

The turtle simulator turtlesim is a classic teaching tool in ros. Through turtlesim you can quickly get to grips with the core concepts in ros.

- Run the turtle simulator

```
ros2 run turtlesim turtlesim_node
```

- Run the keyboard control node

```
ros2 run turtlesim turtle_teleop_key
```

- View the velocity topic data

```
ros2 topic echo /turtle1/cmd_vel
```